

**Practice Exam B**

45 minutes

Total marks: 31

Mathematics

Paper 1

Non-Calculator

FORMULAE LISTVolume of a sphere: $V = \frac{4}{3}\pi r^3$ Volume of a cone: $V = \frac{1}{3}\pi r^2 h$ Volume of a pyramid: $V = \frac{1}{3}Ah$

1. Evaluate $4\frac{1}{5} - 1\frac{2}{3}$. (2)

2. Expand and simplify $(3x - 5)^2 - 4x(x + 1)$. (3)

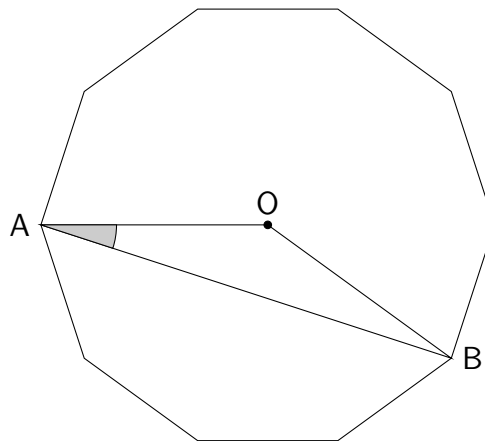
3. In 2023 there were 300 pairs of osprey in Scotland. (3)

This is an increase of 20% from the number of pairs recorded in 2018.

Determine the number of pairs of osprey in Scotland in 2018.

4. Make h the subject of the formula $k = \sqrt[3]{jh} + 4$. (3)

5. The diagram below shows a regular decagon with centre O. (3)



Find the size of the shaded angle OAB.

6. Solve the inequation $2 - \frac{4x}{3} \leq 10$. (3)

7. Determine the gradient of the line with equation $3x + 6y - 4 = 0$. (2)

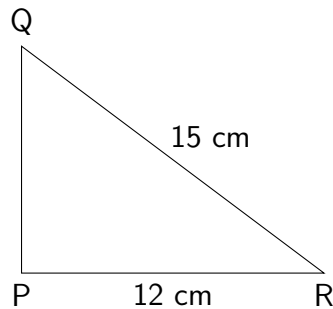
8. A gardener records the mass of a sample of ten apples produced by the tree in her garden. (3)

The results, in grams, are as follows:

203 182 197 146 152 174 135 199 184 171

Find the median and interquartile range of the mass of the apples.

9. Triangle PQR, shown in the **sketch** below, has a perimeter of 36 centimetres. (3)



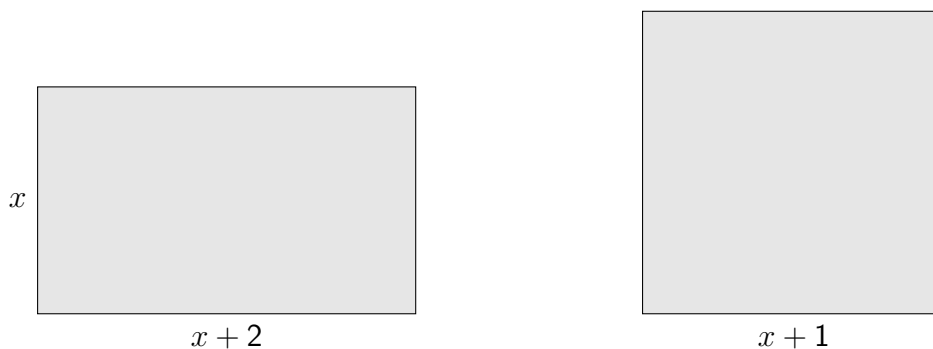
Determine whether triangle PQR is right-angled. **Justify your answer clearly.**

10. Factorise:

(a) $8x^2 - 10x$ (1)

(b) $49 - x^2$ (1)

11. The diagram below shows a rectangle and a square, with lengths shown in centimetres.



The **total area** of the rectangle and square **together** is 31 square centimetres.

(a) Show that $x^2 + 2x - 15 = 0$. (2)

(b) Hence find the value of x . (2)

HINTS: Practice Exam B Paper 1

1. When adding or subtracting fractions, the **denominators** must be the same.
2. To **square** a bracket, first write it out twice: $(3x - 5)^2 = (3x - 5)(3x - 5)$
3. If the number has **increased by 20%** then it is now at **120%** compared to the original...
4. Subtract 4 from each side first, then consider **how to remove the cube root**...
5. One option is to begin by calculating the **central angle** of a regular decagon. **How many** central angles make up angle AOB? What kind of a triangle is AOB?
6. To deal with a fraction in an equation, a common approach is to **multiply both sides by the denominator**.
7. Start by **changing the subject** of the equation to y .
8. Begin by putting the values **in order**...
9. What is the length of the **missing side**, PQ?
10. When factorising look for **common factors**, a **difference of two squares** and **quadratic trinomials / double brackets**.
11. For part (a), find an expression for the area of each shape. Even if you don't manage to complete part (a), you can still **solve the equation to complete part (b)**.

ANSWERS: Practice Exam B Paper 1

1. $2\frac{8}{15}$ **or** $\frac{38}{15}$
2. $5x^2 - 34x + 25$
3. 250 pairs
4. $h = \frac{(k-4)^3}{j}$
5. 18 degrees
6. $-6 \leq x$ **or** $x \geq -6$
7. $-\frac{1}{2}$
8. median = 178 and IQR = 45
9. Triangle PQR is right-angled **with suitable working**
10. (a) $2x(4x - 5)$
(b) $(7 + x)(7 - x)$
11. (a) Clear working leading to $x^2 + 2x - 15 = 0$
(b) $x = 3$